MORADABAD INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING
CLASS TEST-1 (3rd SEMESTER)
Session 2025-26

Subject: Digital Electronics | **Sub Code: BOE310**
MM: 20 | **Duration: 1 HR**
Branch: CSE | **Sec: A, B, C**

faiz

A-10
B-11
C-12
D-13

Q.NO.	1	2	3	4	5	6
CO	1	1	1	3	2	2
Bloom's Level	K1	K2	K2	K3	K6	K6

CO1: Apply concepts of Digital Binary System and implementation of Gates.
CO2: Analyze and design of Combinational logic circuits
CO3: Analyze and design of Sequential logic circuits with their applications.

Note: 1) **This paper contains two sections A and B.**
2) **All sections are compulsory.**

105 138

Section A (08 Marks)

32 16 8 4 2 1
1 0 1 1 1 0
1 0 0 1 0 1

1. Represent the unsigned decimal numbers 791 and 658 in BCD, and then show the steps necessary to form their sum. [2]
2. (i) Convert $(A5D8)_{16}$ into BCD number. [3×1=3]
   (ii) Convert $(46)_{10}$ to Gray Code.
   (iii) Represent **−37** in **8-bit 2's complement.**
3. Simplify using **K-map** and implement with **NAND gates**:
   $F(A,B,C,D) = \Sigma m(0,1,2,5,8,9,10,13)$ [3]

Section B (12 Marks)

2 | 46
2 | 23 0
2 | 11 1
2 | 5 1
2 | 2 1
1

4. Derive the characteristic equation of SR flip flop with block diagram. [4]
5. Design a full adder by half adder and obtain the required expression for sum and carry. [4]
6. Design a combinational circuit that converts a four-bit binary number to a four-bit gray code. [4]